Most real-world Java applications would use a dependency injection library, such as Spring or Guice, to manage dependencies between the modules in a cleaner manner. As you saw earlier, Java-based BDD tools like Cucumber and JBehave provide integration with the main dependency-management libraries, allowing you to automatically configure and inject other production service classes into the service you're testing.

9.3.3 Testing the service layer

Many, if not most, modern applications use some kind of service-based architecture. A service is a well-defined piece of functionality that can be called by other components or applications. Services generally act as facades, gateways, or aggregators for lower-level components (see figure 9.8). In many applications, the UI layer can't invoke lower-level business or database components directly, but instead must use one of the services available in the service layer.

Such services can be incorporated within a larger application, or they may take the form of discrete services that can be deployed and redeployed individually as needed and can be reused across multiple applications. This second approach is what's generally understood by the term *Service Oriented Architecture* (SOA). In both cases, a well-designed service architecture allows services to be combined and reused in different parts of an application, or by different applications, to build and deliver new business functionality more quickly and more reliably.

The service layer plays a key role in good system architecture. You've already seen how many web applications are now built around sophisticated client applications, such as modern JavaScript-based single-page applications using frameworks like AngularJS and Backbone. And as you've seen, these client applications have their own complex business logic and behavior, and they invoke web services to obtain the data they need. Many mobile applications are implemented using a similar approach, with a native client application calling remote services.

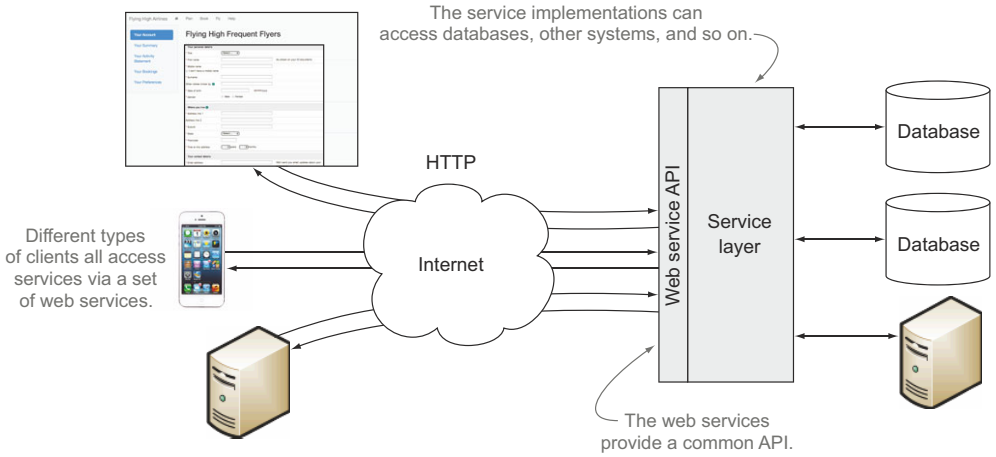


Figure 9.8 An example of an architecture using a service layer